

Multiple Choice

1. **Answer:** D
Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above
Note: Correct option: D - All of the above
2. **Answer:** A
Options: A) PCA; B) Decision Tree; C) Linear Regression; D) Naive Bayesian
Note: Correct option: A - PCA
3. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
4. **Answer:** D
Options: A) 3; B) 4; C) 7; D) 15
Note: Correct option: D - 15
5. **Answer:** B
Options: A) Attributes are equally important.; B) Attributes are statistically dependent of one another given the class value.; C) Attributes are statistically independent of one another given the class value.; D) Attributes can be nominal or numeric
Note: Correct option: B - Attributes are statistically dependent of one another given the class value.

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'The use of prior distributions on the parameters in a probabilistic model.'
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'lower'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, False'.

Long Form Response

11. **Answer:** `def max_path_sum(tri, m, n): | return tri[0][0]`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def text_match_three(text): | return 'Found a match!'`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key